

How does unlabeled data improve generalization in self-training?

Shuai Zhang+, ICLR 2022

발표자: 송헌 (songheony@gmail.com)

펀디멘탈팀: 김동현, 김채현, 박종익, 양현모, 오대환, 이근배, 이재윤

- Under one-hidden-layer NN, it quantify the impact of labeled and unlabeled data on the generalization of the model in iterative self-training on regression task.
- Based on that, it explains how iterative self-training works well.
- Moreover, it suggests that more data, better performance.

Self-training

- Iterative self-training is summarized as follows:
 - a. Initialize iteration $\ell=0$ and obtain a model $g(W^{(\ell)})$ as the teacher using labeled data only.
 - b. Use the teacher model to obtain pseudo labels of unlabeled data
 - c. Train the neural network $g(W^{(\ell+1)})$ by minimizing the empirical risk.
 - d. Use $g(W^{(\ell+1)})$ as the current teacher model and go back to step b.
- Given labeled dataset $D=\{x_n, y_n\}_{n=1}^N$ and unlabeled dataset $D^{\sim}=\{x_m, y_m\}_{m=1}^M$, the **empirical risk** can be defined as follows:

$$\hat{f}_{D, \tilde{D}}(W) = \frac{\lambda}{2N} \sum_{n=1}^N (y_n - g(W; x_n))^2 + \frac{\tilde{\lambda}}{2M} \sum_{m=1}^M (\tilde{y}_m - g(W; x_m))^2$$

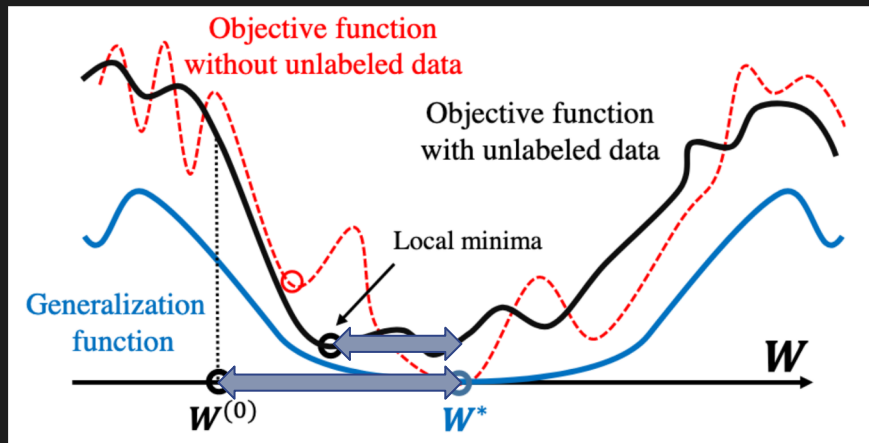
where $\lambda + \tilde{\lambda} = 1$.

Generalization function

- Given unknown ground-truth model $g(W^*)$, generalization function is defined as:

$$I(g(W)) = \mathbb{E} \left[(y - g(W; x))^2 \right] = \mathbb{E} \left[(g(W^*; x) - g(W; x))^2 \right]$$

- Authors does not directly analyze $I(g(W))$ but analyze the distance $\|W - W^*\|_F$, and they shows that $I(g(W))$ is linear in $\|W - W^*\|_F$ numerically.



Q&A

Proof of the main theorem

- Zhong 2017 studies about one-hidden-layer neural networks.
- Assuming data is drawn from standard Gaussian distribution,
- First, they shows $I(g(\mathbf{W}))$ is locally convex near $\mathbf{W}^*$.
- Second, if the number of data is sufficiently large, N^* , the empirical risk can approximate $I(g(\mathbf{W}))$ well in the neighborhood of $\mathbf{W}^*$.
- Third, the proposed initialization method makes $\mathbf{W}^0$ be in the local convex area.
- Consequently, supervised learning can return the ground-truth model $g(\mathbf{W}^*)$.
- Differences between papers: 1) the number of labeled samples is less than N^* and 2) $\mathbf{W}^*$ is not the minimizer of empirical risk in Zhang 2022.

Finite sample guarantees

- Suppose the iteration number is sufficiently large and

$$N^*/4 \leq N, \hat{\lambda} \leq \sqrt{N/N^*}, O(1 - \hat{\lambda}) \leq M$$

where $\hat{\lambda}$ is defined by λ and $\lambda^\sim$ and increasing function of λ .

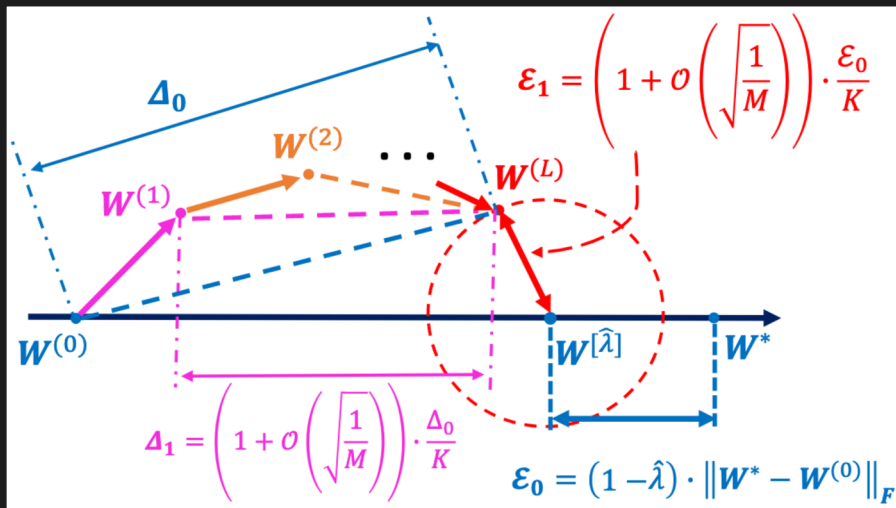
- By minimizing empirical risk, the trained model satisfies

$$\left\| W^{(L)} - W^* \right\|_F \leq O \left((1 - \hat{\lambda}) / \sqrt{M} \right) \left\| W^{(0)} - W^* \right\|_F$$

- When $\hat{\lambda}$ is increased, 1) the required number of unlabeled data is reduced
2) the new weight $W^{(L)}$ becomes to closer to W^* .

Highlights

- The convergence rate is proportional to $1 / \sqrt{M}$.
- The iterative self-training can return a model in the neighborhood of $\mathbf{W}^{[\hat{\lambda}]}$ where $\mathbf{W}^{[\hat{\lambda}]} = \hat{\lambda} \times \mathbf{W}^* + (1 - \hat{\lambda}) \times \mathbf{W}^{(0)}$.
- The distance between $\mathbf{W}^{(L)}$ and $\mathbf{W}^{[\hat{\lambda}]}$ scales in the order of $1 / \sqrt{M}$.



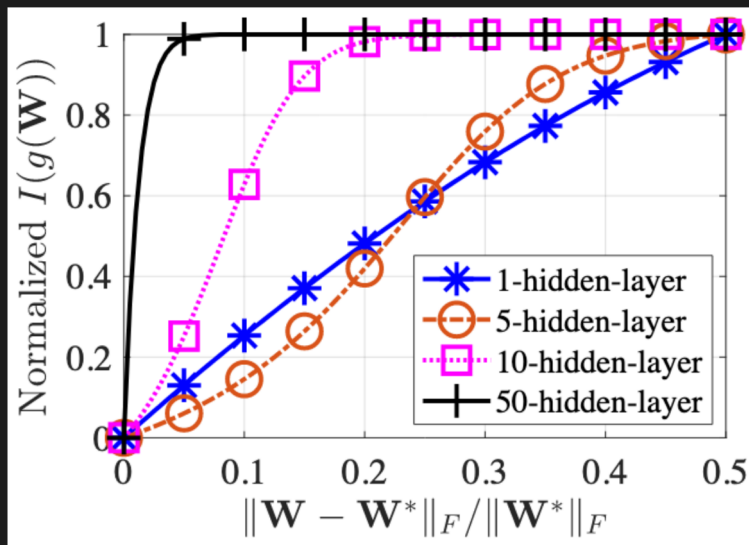
Q&A

Synthetic data experiments

- A ground-truth NN with 10 hidden neurons is generated.
- The labeled and unlabeled samples are drawn from $\mathcal{N}(0, I)$.
- The input dimension is set to 50.
- The value of λ is properly selected to meet assumption.
- Self-training terminates if $\|W - W^*\|_F$ becomes small enough until 1000 iteration.

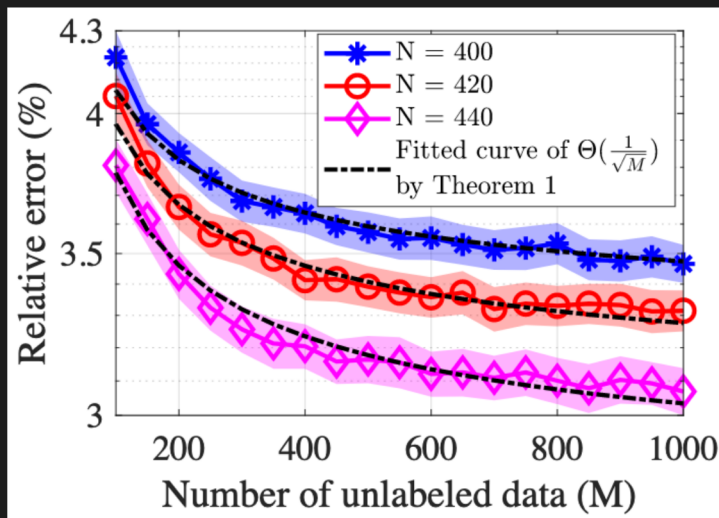
$I(g(\mathbf{W}))$ proportional to $\|\mathbf{W} - \mathbf{W}^*\|$

- The $I(g(\mathbf{W}))$ is plotted against the distance to the ground-truth weight.
- For one-hidden-layer, $I(g(\mathbf{W}))$ is almost linear in $\|\mathbf{W} - \mathbf{W}^*\|_F$ in a large region.
- When the number of hidden layers increases, this region decreases, but the linear dependence still holds locally.



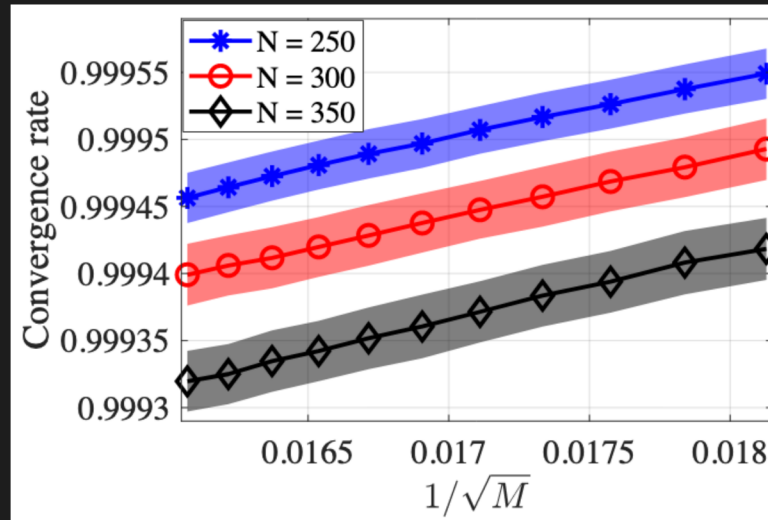
$\|W - W^*\|$ as a linear function of $1 / \sqrt{M}$

- Relative error ($\|W - W^*\|_F / \|W^*\|_F$) is plotted changing M .
- The relative error decreases when either M or N increase.
- Dash-dotted lines represent the best fitting of the linear functions of $1 / \sqrt{M}$.
- Therefore, the relative error is a linear function of $1 / \sqrt{M}$.



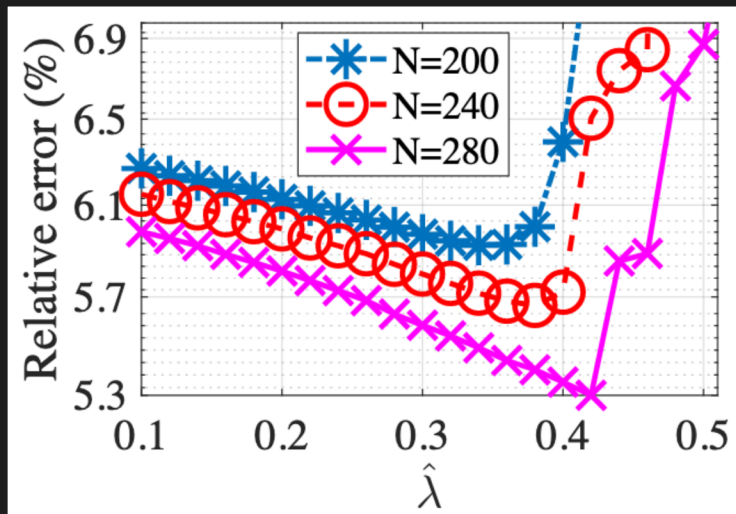
Convergence rate as a linear function of $1 / \sqrt{M}$

- The convergence rate is plotted changing M .
- The convergence rate is a linear function of $1 / \sqrt{M}$.
- When M increases, the convergence rate is improved.



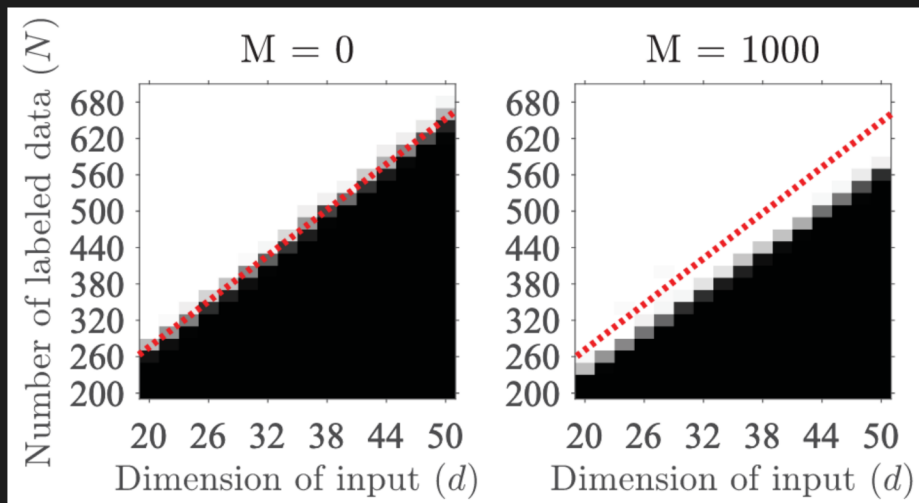
Relative error is improved as a linear function of $\lambda^\wedge$

- The relative error is plotted against $\lambda^\wedge$.
- The relative error decrease almost linearly when $\lambda^\wedge$ increases.
- Moreover, when $\lambda^\wedge$ exceeds a certain threshold positively correlated with N , The relative error increases rather than decreases.



Unlabeled data reduce the sample complexity

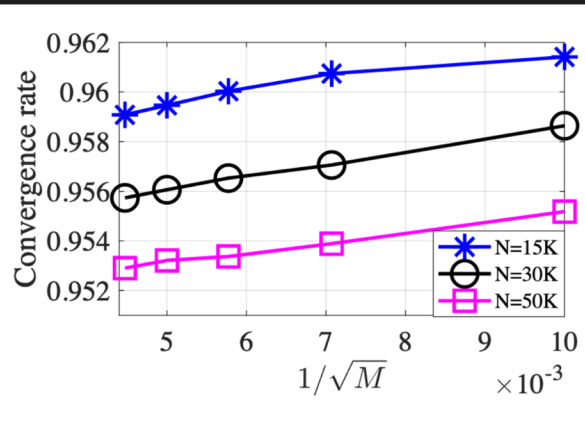
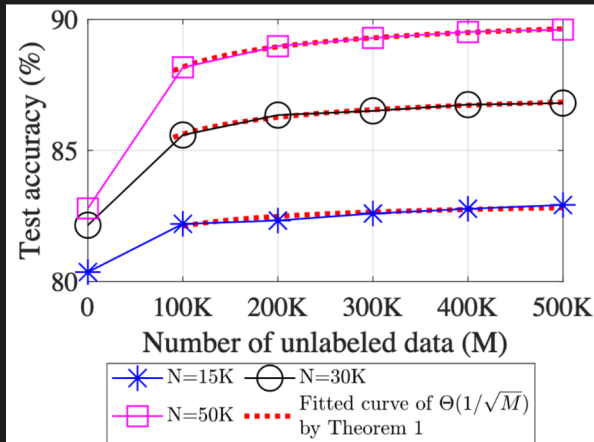
- For every pair of d and N , 100 independent trials are conducted.
- The white blocks correspond to low average relative error.
- The required number of N is linear in d .
- Moreover, with unlabeled data, the required sample complexity of N is reduced.



Q&A

Image classification on real-world dataset

- ResNet is trained on labeled CIFAR-10 and unlabeled 500k images.
- λ and $\lambda^\sim$ are selected as $N/(M+N)$ and $M/(M+N)$, respectively.
- The test accuracy is improved by using unlabeled data, and the empirical evaluations match the theoretical predictions.
- Moreover, the convergence rate is almost a linear function of $1/\sqrt{M}$.



Conclusion

- The authors showed that the improved generalization error and convergence rate is a linear function of $1 / \sqrt{M}$, theoretically.
- Moreover, they demonstrated the unlabeled data improved generalization as they expected empirically through the experiments.
- However, there are several limitations that
 - Data is assumed to be drawn from standard Gaussian distribution
 - Not multi-layer NN but two-layer NN
 - Not classification but regression